

## **APPENDIX B**

# forecasting the markets? A small digression

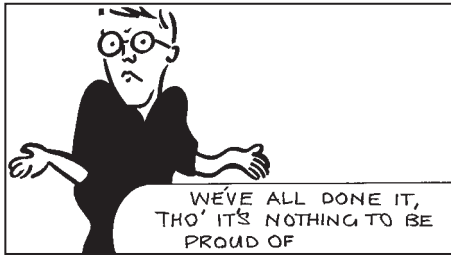


### **The aim of this Appendix...**

...is to explain ways in which people supposedly predict future movements in the financial markets. There is little scientific evidence that these methods work in practice but you, a future bond trader perhaps, must know about such matters and eventually decide for yourself.

### **In this Appendix...**

- some of the commonly used technical methods for predicting market direction
- some modern approaches to modeling markets and their microstructure



## B.1 INTRODUCTION

People have been making predictions about the future since the dawn of time. And predicting the future of the financial markets has been especially popular. Despite the claims of many ‘legendary’ investors it is not clear whether there is any validity in any of the methods they use, or whether the claims are examples of survivor bias. The big losers tend to keep quiet.

In this appendix we look at some of the traditional methods for determining trends, technical analysis, and also some of the more recent methods, often emanating from physics. I won’t be describing some of the more dubious ideas, such as astrology, but then we Scorpions tend to be skeptical.

In the book generally, I’m taking the view that the markets are best modeled via probabilities. This appendix is very much a digression from the main thrust of the book.

## B.2 TECHNICAL ANALYSIS

**Technical analysis** is a way of predicting future price movements based only on observing the past history of prices. This price history may also include other quantities such as volume of trade. These methods contrast with **fundamental analysis** in which prediction is made based on an examination of the factors underlying the stock or other instrument. This may include general economic or political analysis, or analysis of factors specific to the stock, such as the effect of global warming on snowfall in the Alps, if one is concerned with a travel company. In practice, most traders will use a combination of both technical and fundamental analysis.

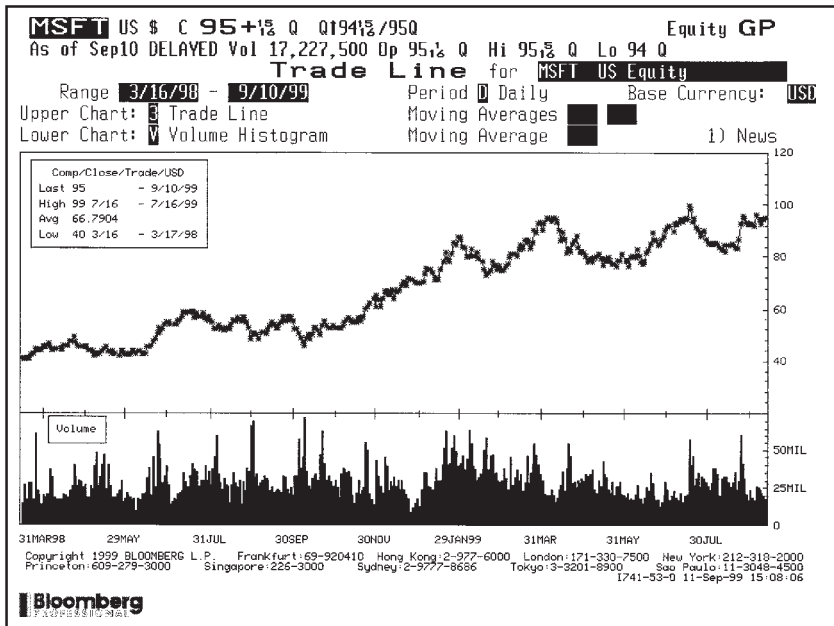
Technical analysis is also called **charting** because the graphical representation of prices, etc. plays an important part. Technical analysis is thought to be particularly good for timing market moves; fundamental analysis may get the direction right, but not necessarily when the move will happen.

### B.2.1 Plotting

The simplest chart types just join together the prices from one day to the next, with time along the horizontal axis. These are the sort of plots we have seen throughout this book. Sometimes a logarithmic scale is used for the vertical price axis to represent return rather than absolute level. Later on we’ll see some more complicated types of plotting. Sometimes you will see trading volume on the same graph, this is also used for prediction but I won’t go into any details here, see Figure B.1.

### B.2.2 Support and resistance

**Resistance** is a price level which an asset seems to have difficulty rising above. This may be a previously realized highest value, or it may be a psychologically important (round) number. **Support** is a level below which an asset price seems to be reluctant to fall.



**Figure B.1** Price and volume. Source: Bloomberg L.P.

There may be sufficient demand at this low price to stop it falling any further. Examples of support and resistance are shown in Figure B.2.

When a support or resistance level finally breaks it is said to do so quite dramatically.

### B.2.3 Trendlines

Similar to support and resistance are **trendlines**. These are formed by joining together successive peaks and/or troughs in the price history to form a rising or falling support or resistance level. An example is shown in Figure B.3.

### B.2.4 Moving averages

**Moving averages** are calculated in many ways. Different time windows can be used, or even exponentially weighted averages can be calculated. Moving averages are supposed to distill out the basic trend in a price by smoothing the random noise.

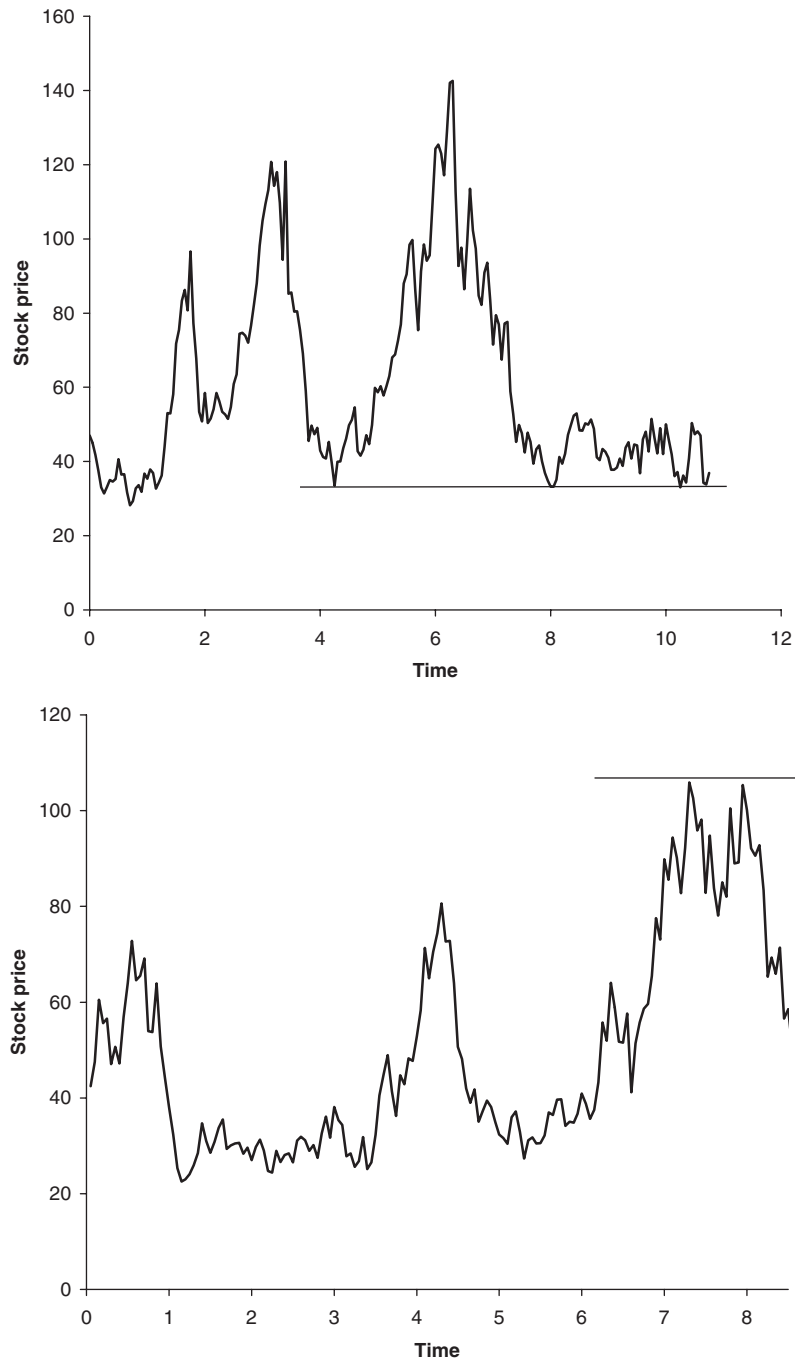
Sometimes two moving averages are calculated, say a 10-day and a 250-day average. The crossing of these two would signify a change in the underlying trend and a time to buy or sell.

Although I'm not the greatest fan of technical analysis, there is some evidence that there may be predictive power in moving averages.

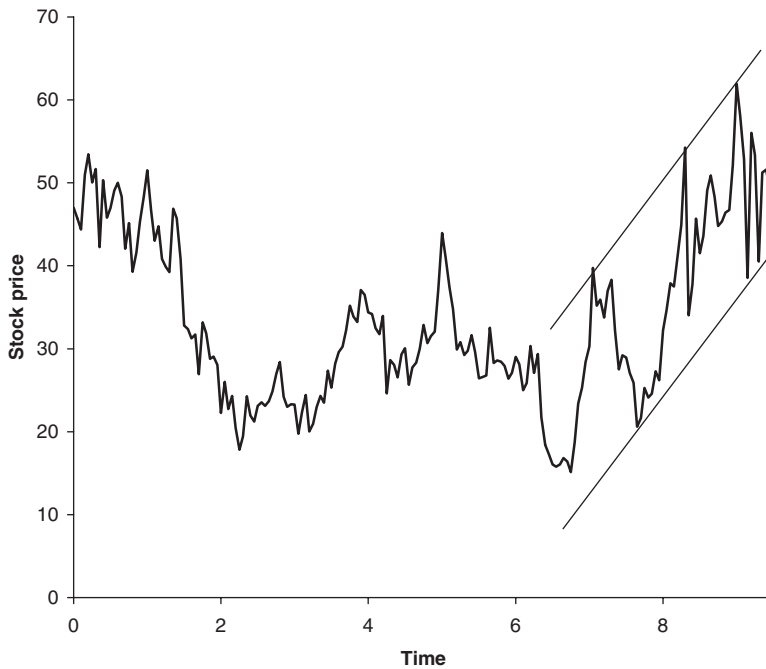
Figure B.4 shows a Bloomberg screen with Microsoft share price, five- and 15-day moving averages.



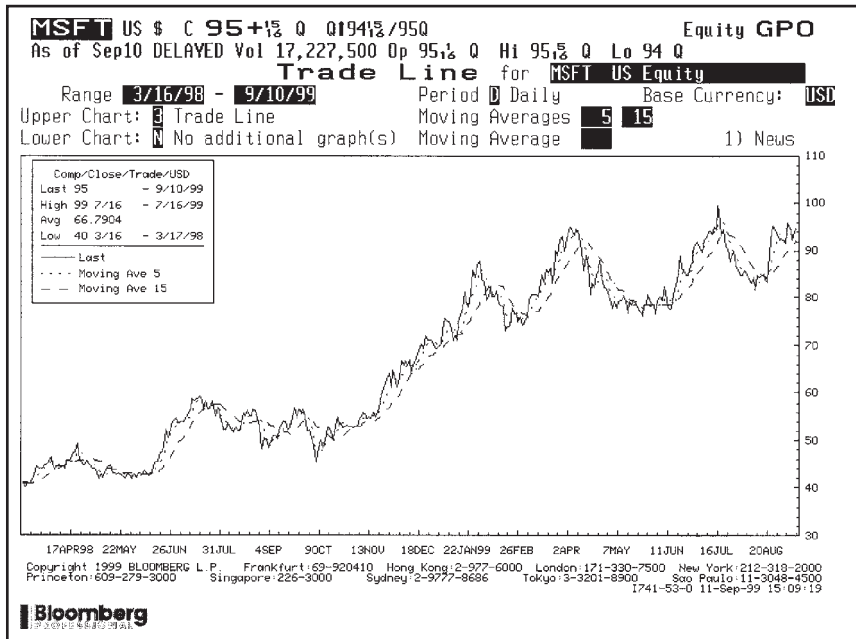
Moving averages may actually be useful



**Figure B.2** Support and resistance.



**Figure B.3** A trending stock.



**Figure B.4** Two moving averages. Source: Bloomberg L.P.



RSI

**B.2.5** Relative strength

The **relative strength index** is the percentage of up moves in the last  $N$  days. A number higher than 70% is said to be overbought and is therefore likely to fall, and below 30% is said to be oversold and should rise.

**B.2.6** Oscillators

An **oscillator** is another indicator of over/underbought conditions. One way of calculating it is as follows.

Define  $k$  by

$$100 \times \frac{\text{Current close} - \text{lowest over } n \text{ periods}}{\text{Highest over } n \text{ periods} - \text{lowest over } n \text{ periods}}$$

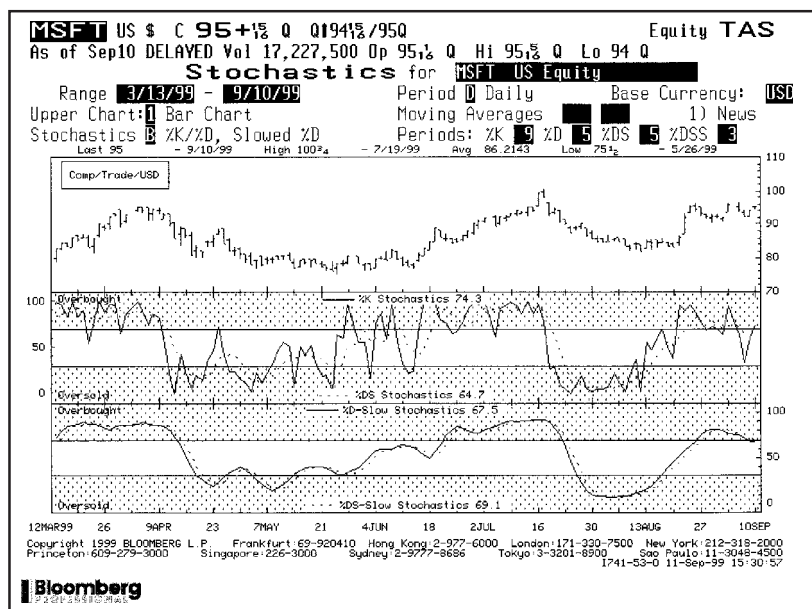
Now take a moving average of the last three days, say. This average is plotted against time and any move outside the range 30–70% could be an indication of a move in the asset. See Figure B.5.



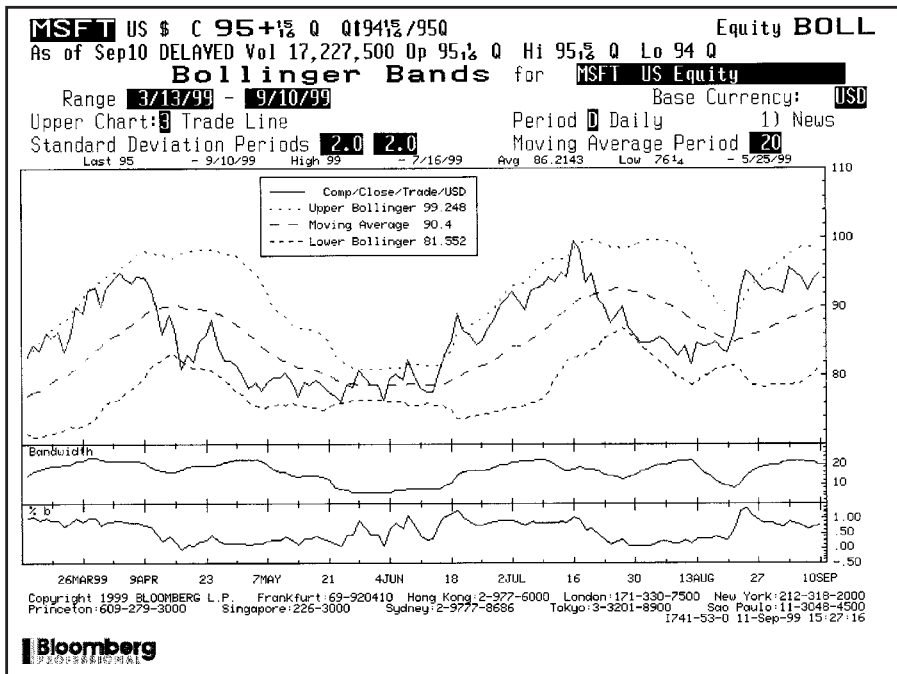
Oscillators

**B.2.7** Bollinger bands

**Bollinger bands** are plots of a specified number of standard deviations above and below a specified moving average, see Figure B.6.



**Figure B.5** Oscillator. Source: Bloomberg L.P.



**Figure B.6** Bollinger bands. Source: Bloomberg L.P.

### B.2.8 Miscellaneous patterns

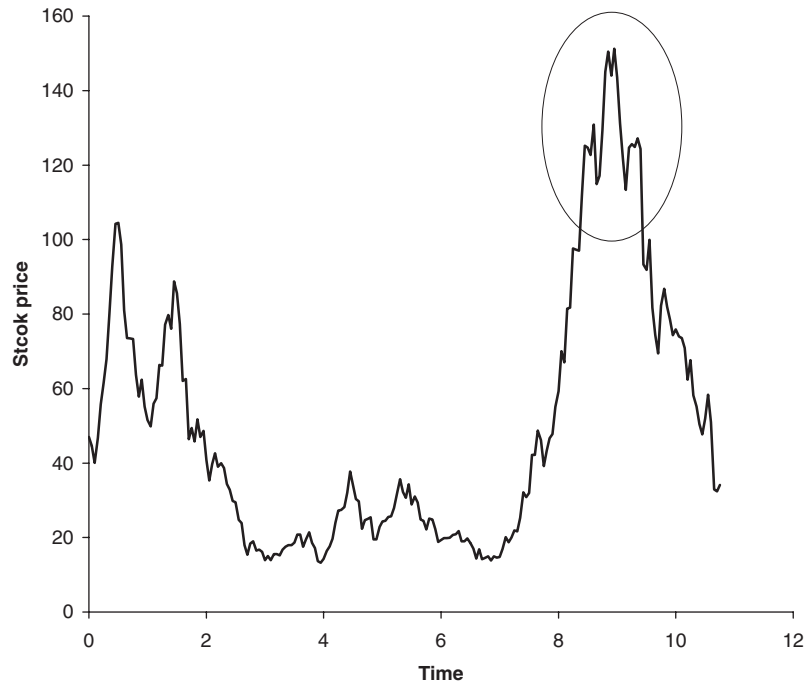
As well as the 'quantitative' side of charting there is also the 'artistic' side. Practitioners say that certain patterns anticipate certain future moves. It's rather like your grandmother reading tea leaves.

**Head and shoulders** is a common pattern and is best described with reference to Figure B.7. There are a left and a right shoulder with the head rising above. Following on from the right shoulder should be a dramatic decline in the asset price.

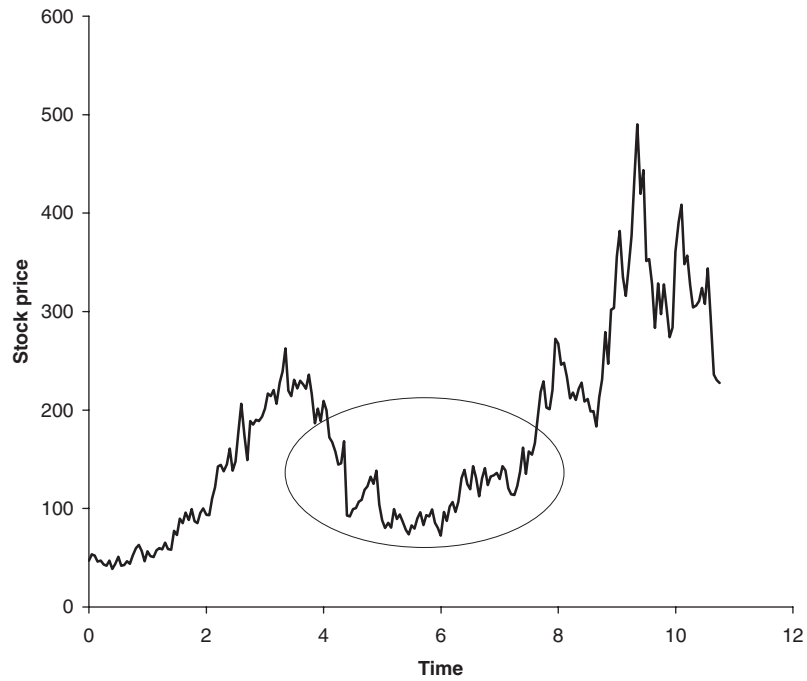
This pattern is supposed to be one of the most reliable predictors. It is also seen in an upside-down formation.

**Saucer tops and bottoms** are also known as **rounding tops** and **bottoms**. They are the result of a gradual change in supply and demand. The shape is generally fairly symmetrical as the prices rise and fall. These patterns are quite rare. They contain no information about the strength of the new trend. See Figure B.8.

**Double and triple tops and bottoms** are quite rare patterns, the triple being even rarer than the double. The double top looks like an 'M' and a double bottom like a 'W.' The triple top is similar but with three peaks, as shown in Figure B.9. The key point about the peaks and troughs is that they should all be at approximately the same level.

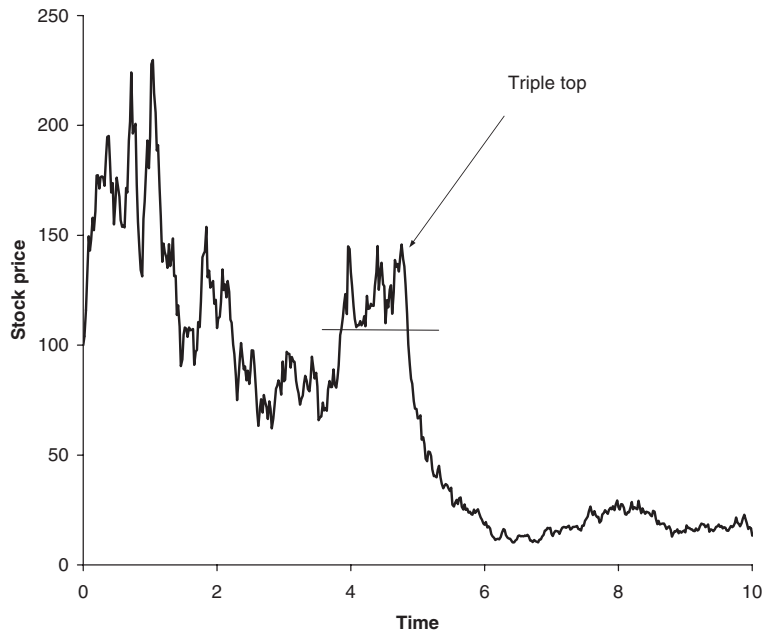


**Figure B.7** Head and shoulders.



**Figure B.8** Saucer bottom.





**Figure B.9** A triple top.

### **B.2.9** Japanese candlesticks

**Japanese candlesticks** contain more information than the simple plots described so far. They record the opening and closing prices as well as the day's high and low. A rectangle is drawn extending from the close to the open, and is colored white if close is above open and black if close is below open. The high-low range is marked by a continuous line.

Certain combinations of candlesticks, appearing consecutively, have special meanings and names like 'Hanging Man' and 'Upside Gap Two Crows.' See Figure B.10 for the two types of candlestick and see Figure B.11 for candlesticks in action. On this chart are shown 'HR' = Bearish Harami, 'D' = Doji (representing indecision), 'BH' = Bullish Harami, 'EL' = Bearish Engulfing Line, and 'H' = Hanging Man (representing reversal after a trend).

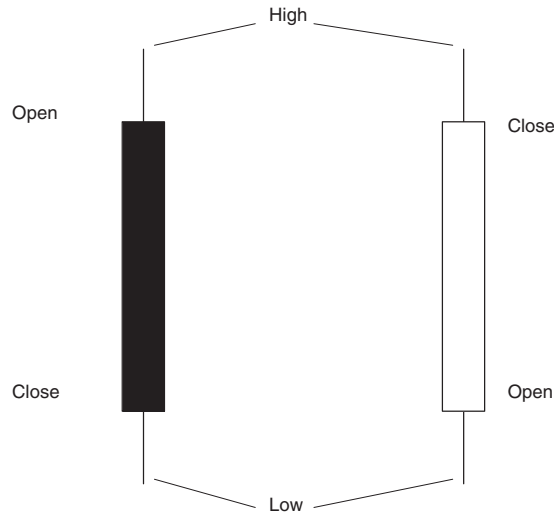
Figure B.12 shows some of the possible candlestick shapes and their interpretation.



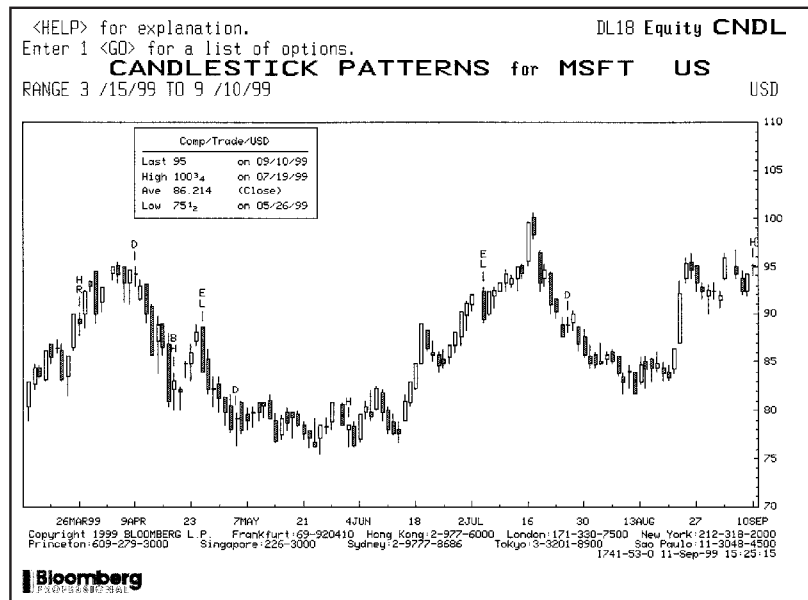
Excel has options for plotting candlesticks

### **B.2.10** Point and figure charts

**Point and figure charts** are different from the charts described above in that they do not have any explicit timescale on the horizontal axis. Figure B.13 is an example of a point and figure chart. Each box on the chart represents a prespecified asset price move. The boxes are a way of discretizing asset price moves, instead of discretizing in time. For each consecutive asset price rise of the box size draw an 'X' in the box, in a rising column,



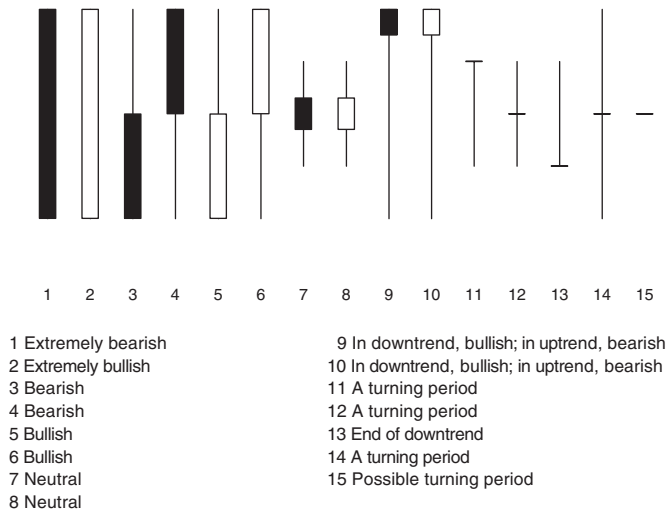
**Figure B.10** Japanese candlesticks.



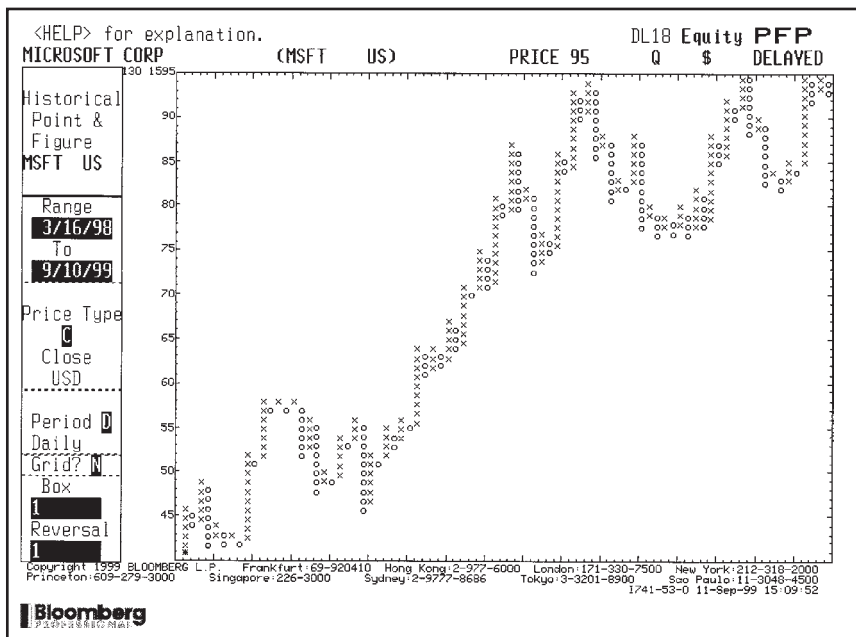
**Figure B.11** A candlestick chart. Source: Bloomberg L.P.

one above the other. When this uptrend finishes, and the asset falls, start putting 'O' in a descending column, to the right of the previous rising Xs.

- A long column of Xs denotes demand exceeding supply.
- A long column of Os denotes supply exceeding demand.
- Short up and down columns denote a balance of supply and demand.



**Figure B.12** The meanings of the various candlesticks.



**Figure B.13** A point and figure chart of Microsoft. Source: Bloomberg L.P.

### B.3 WAVE THEORY

As well as plotting and spotting trends in price movements there have been some theories for price prediction based on market cycles or waves. Below, I briefly mention a couple.

**B.3.1** Elliott waves and Fibonacci numbers

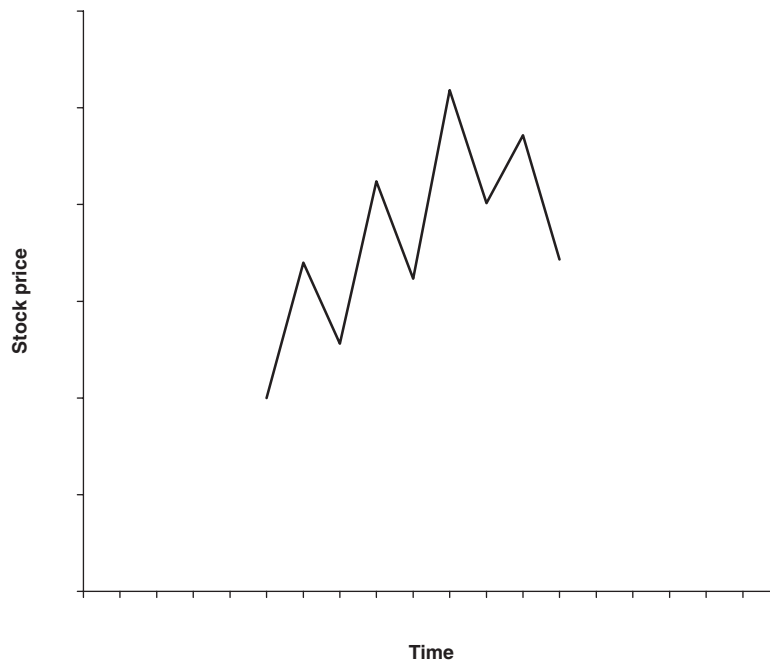
Ralph N. Elliott observed repetitive patterns, waves or cycles in prices. Roughly speaking, there are supposed to be five points in a bullish wave and then three in a bearish one. See Figure B.14. Within this **Elliott wave theory** there is also supposed to be some predictive ability in terms of the sizes of the peaks in each wave. For some reason, the ratio of peaks in a trend are supposed to be fairly constant; the ratio of second peak to first should be approximately 1.618 and of the third to the second 2.618. Unfortunately, the number 1.618 is approximately the **Golden ratio** of the ancient Greeks;  $\frac{1}{2}(\sqrt{5} + 1)$ . It is also the ratio of successive numbers in the **Fibonacci series** given by  $a_n = a_{n-1} + a_{n-2}$  for large  $n$ . I say, unfortunately, because people extrapolate wildly from this. And if it's a coincidence then . . . Figure B.15 shows the key levels coming from the Fibonacci series.

**B.3.2** Gann charts

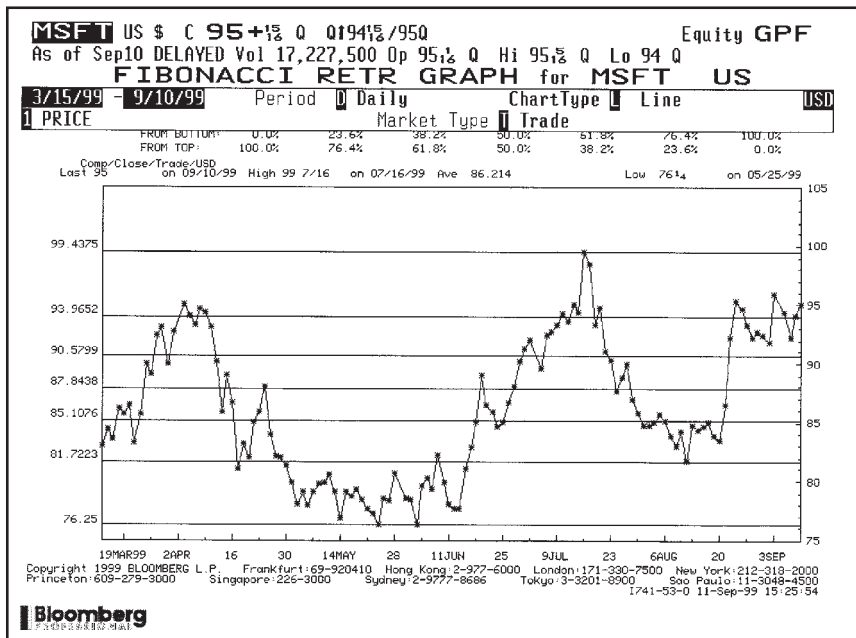
Figure B.16 shows a Gann chart. The lines all have slopes which are fractions of the slope of the lowest line. Need I say more?

**B.4 OTHER ANALYTICS**

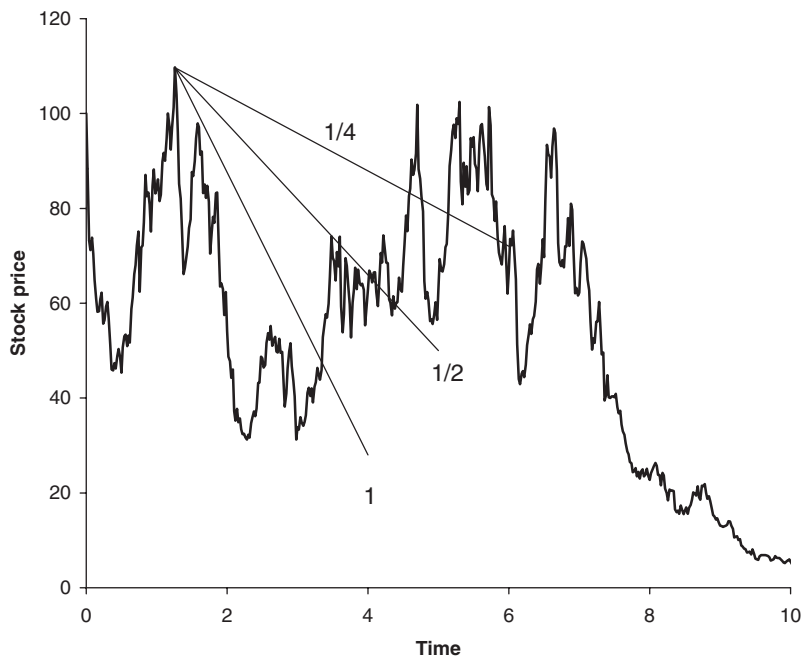
There's an almost endless number of ways that chartists analyze data. I'll mention just a couple more before moving on.



**Figure B.14** Elliot waves.



**Figure B.15** Fibonacci lines. Source: Bloomberg L.P.



**Figure B.16** Gann charts.

**Volume** is simply the number of contracts traded in a given period. A rising price and high volume means a strong, upwardly trending market. But a rising price with low volume could be a sign that the market is about to turn.

**Open interest** is the number of still outstanding futures contracts, those which have not been closed out. Because there are equal numbers of buyers and sellers, open interest does not necessarily give any directional info, but an increase in open interest can mean that an existing trend is strong.

## B.5 **MARKET MICROSTRUCTURE MODELING**

The financial markets are made up of many types of players. There are the ‘producers’ who manufacture or produce or sell various goods to who may be involved in the financial markets for hedging. There are the ‘speculators’ who try to spot trends in the market, to exploit them and make money. These speculators may be using technical analysis methods, such as those described above, or fundamental analysis, whereby they examine the records and future plans of firms to determine whether stocks are under- or overpriced. Almost all traders use technical analysis at some time. Then there are the market makers who buy and sell financial instruments, holding them for a very short time, often seconds, and profit on bid-offer spreads.

There have been many attempts to model the interaction of these agents, sometimes in a game theoretic way, to try to model the asset price movements that in this book we have taken for granted. For example, can the dynamics induced by the actions of a combination of these three types of agent result in Brownian motion and lognormal random walks?

Below are just a very few examples of work in this area.

### **B.5.1** Effect of demand on price

Buying and selling assets moves their prices. Market makers respond to demand by increasing price, and reduce prices when the market is selling. If one can model the relationship between demand and price then it should be possible to analyze the effect that various types of technical trading rule have on the evolution of prices. And eventually to model the dynamics of prices.

A common starting point is to assume that there are two types of trader and one market maker. One trader follows a technical trading rule such as watching a moving average and the other is a **noise trader** who randomly buys or sells.

Interesting results follow from such models. For example:

- trend followers can induce patterns in asset price time series;
- these artificially induced patterns can only be exploited for gain by someone following a suitably different trend;
- the more people following the same trend as you, the more money you will lose.

There are good reasons for there being genuine trends in the market: there is a slow diffusion of information from the knowledgeable to the less knowledgeable; the piece-by-piece secret acquisition of a company will gradually move a stock price upwards.

On the other hand, if there is no genuine reason for a trend, if it is simply a case of trend followers begetting a trend, then it may be beneficial to be a contrarian.

### **B.5.2** Combining market microstructure and option theory

Arbitrage does exist, many people make money from its existence. Yet the action of arbitragers will, via a demand/price relationship, remove the arbitrage. But there will be a timescale associated with this removal. What is the optimal way to exploit the arbitrage opportunity while knowing that your actions will to some extent be self-defeating?

### **B.5.3** Imitation

Another approach to market microstructure modeling is based on the true observation that people copy each other. In these models there are a number of traders who act partly in response to private information about a stock, partly randomly as noise traders, and partly to imitate their nearest neighbors. These models can result in market bubbles or market crashes.

## **B.6 CRISIS PREDICTION**

There has been some work on analyzing data over various timescales to determine the likelihood of a market crash. Some ideas from earthquake modeling have been used to derive a 'Richter'-like measure of market moves. Of course, an effective predictor of market crashes could either

- increase the chance or size of a crash as everyone panics or
- reduce the chance or size of the crash since everyone gets advance warning and can calmly and logically act accordingly.

## **B.7 SUMMARY**

I started out in finance many years ago plotting all of the technical indicators. I was not very successful at it. I could only get directions right for those assets with obvious seasonality effects, such as some commodities.

There is only one technical indicator that I believe in. There is definitely a strong correlation between hemlines and the state of the economy. The shorter the skirts, the better the economy.

## **FURTHER READING**

- The book on technical analysis written by the news agency Reuters (1999) is excellent, as is Meyers (1994).
- Farmer & Joshi (2000) discusses and models trend following, and the creation of trends. They also demonstrate properties of the relationship between demand and price that prevent arbitrage.
- Bhamra (2000) has worked on imitation in financial markets.
- Olsen & Associates [www.olsen.ch](http://www.olsen.ch) are currently working in the area of crisis modeling and prediction.
- Johnson *et al.* (1999) models self-organized segregation of traders, and concludes that cautious traders perform poorly.
- The above is only a brief description of a very few examples from an expanding field. See O'Hara (1995) for a wide-ranging discussion of market microstructure models.
- Bernstein (1998) has a whole chapter on the Golden ratio.
- Elton & Gruber (1995) describe the efficient market hypothesis and criticize technical analysis.
- Prast (2000a, b) discusses 'herding' in the financial markets.